

## EXECUTIVE SUMMARY

This project demonstrates a full cybersecurity incident lifecycle, including attack simulation, detection, triage, and response. A vulnerable system (Metasploitable2) was targeted using Kali Linux with Metasploit, exploiting the VSFTPD 2.3.4 backdoor vulnerability. The attack originated from IP address 192.168.56.102 and successfully established a command shell session on the target system (192.168.56.103).

A Wazuh SIEM environment was configured on an Ubuntu machine (192.168.56.5) to monitor system logs and detect suspicious activity. Log monitoring was enabled using syslog integration, allowing visibility into attack-related events. Alerts were analyzed to identify indicators of compromise and mapped to the MITRE ATT&CK framework (T1190 – Exploit Public-Facing Application).

As part of incident response, containment actions were simulated by isolating the compromised system and planning IP blocking using CrowdSec. Network connectivity tests confirmed the effectiveness of isolation measures.

This project highlights the importance of centralized logging, real-time monitoring, and structured incident response in a Security Operations Center (SOC) environment.

## TIMELINE TABLE

| Timestamp        | Source IP      | Target IP      | Alert Description        | MITRE Technique |
|------------------|----------------|----------------|--------------------------|-----------------|
| 2026-03-29 15:24 | 192.168.56.102 | 192.168.56.103 | VSFTPD exploit attempt   | T1190           |
| 2026-03-29 15:24 | 192.168.56.102 | 192.168.56.103 | Reverse shell opened     | T1059           |
| 2026-03-30 02:00 | 192.168.56.5   | localhost      | Log ingestion configured | T1070           |

## ATTACK SIMULATION

The attack was performed using Kali Linux with Metasploit. The module `exploit/unix/ftp/vsftpd_234_backdoor` was used to exploit the vulnerable VSFTPD service running on Metasploitable2. After setting the target IP, the exploit successfully created a command shell session, confirming remote code execution.

## DETECTION & TRIAGE

Wazuh SIEM was configured on Ubuntu to monitor system logs. Syslog ingestion was enabled to capture events generated during the attack. Logs were analyzed to identify suspicious activity such as unauthorized access attempts and shell execution. Alerts were correlated with MITRE ATT&CK techniques to classify the attack.

## RESPONSE

As part of incident response, containment actions were simulated. The compromised system was logically isolated, and network connectivity was tested using ping commands. Plans were made to block the attacker's IP using CrowdSec. These steps demonstrate basic SOC response procedures to mitigate threats.

## RESPONSE (IMPORTANT)

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## STAKEHOLDER BRIEFING

A simulated cyberattack was conducted to test our security monitoring and response capabilities. The attack successfully exploited a vulnerable service and gained access to a test system. Our monitoring tool (Wazuh) was configured to detect such activities through log analysis. Upon detection, response actions were taken to isolate the affected system and prevent further risk. This exercise highlights the importance of proactive monitoring, timely detection, and structured response in protecting organizational assets. Improvements such as stronger

access controls and automated threat blocking are recommended to enhance overall security posture.